$C_C = 38.0$ J/K

$C_W = 1.6 \times 10^2 \times 4.2$

$= 6.72 \times 10^2$ J/K

$C_\Sigma = 710$ J/K

P18
Q5.

$C_I = 0.44 \times 40$

$= 1.76 \times 10$ J/K

熱 保存則より

$$\underbrace{38(t-15)}_{\text{容器}} + \underbrace{1.6 \times 10^2 \times 4.2 \times (t-15)}_{\text{水}} = \underbrace{40 \times 0.44 \times (90-t)}_{\text{鉄 放出}}$$

$Q = C\Delta t$

$= 710 \times (x - 15)$

$$0.44 = \frac{Q}{m\Delta t} = \frac{Q}{40(90-x)}$$

$17.6(90-x) = 710(x-15)$

$710x + 17.6x = 710 \times 15 + 17.6 \times 90$

$727.6x = 10650 + 1584$

$727.6x = 12234$

$x = 16.814$

$\simeq 16.8$

$\Delta t = 1.8$

問5.

$$-4℃氷 \xrightarrow[Q_a]{} 0℃氷 \xrightarrow[Q_b]{} 0℃水 \xrightarrow{Q_c} t℃水$$

20g　　= g　　= g　　= g

$Q_a = 20 \cdot 2.1 \cdot 40$

$Q_b = 20 \cdot 3.3 \times 10^2$

$Q_c = 20 \cdot 4.2 \cdot t$

$$\sum Q = 168 + 6600 + 84t$$
$$= 80 \cdot 4.2 \cdot (60 - t)$$
$$t = 32℃$$

問6. $-20℃氷 \rightarrow 0℃水$ に要する熱量.

$$-20℃氷 \xrightarrow[Q_a]{} 0℃氷 \xrightarrow[Q_b]{} 0℃水.$$

$$Q_a = 6.0 \times 10^2 \times 2.1 \times 20$$
$$Q_b = 3.3 \times 10^2 \times 6.0 \times 10^2$$
$$Q_\Sigma = 2.232 \times 10^5 \, J.$$

$W = J/s$ だから.

$$t = \frac{J}{W}$$
$$= \frac{2.232 \times 10^{\cancel{5}\,3}}{5.0 \times \cancel{10^2}}$$
$$= 4.464 \times 10^2 \, s$$
$$\fallingdotseq 7.4 \, min$$